Misorientation accumulation along columnar grains in GammaPrint-700 laser powder bed fusion: 3D-EBSD (Lamb et al.) versus SPPARKS

SURGE-kokkos branch, `additive/texture/kk` Moser deck

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Abstract

Lamb et al. measured the starting reference orientation deviation (SROD) of columnar grains in a 3D-EBSD volume of laser powder bed fusion GammaPrint-700. Re-analysing their Fig. 5 shows that SROD is a random-axis rotation walk: $\mathrm{RMS}^2 = Dh$ with $D \approx 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$ against build height, the same D for grains originating inside the volume and for established grains, and a 5/95 band of $(p_{95} - p_5)/\mathrm{mean} \approx 1.4$, the Maxwell width of a three-dimensional walk. SPPARKS `additive/texture` and `additive/texture/kk` now implement that model (`misorientation_diffusion`), and a Moser-source GP700 deck reproduces the experiment: RMS deviation from Lamb's mean 0.49° over 25–240 µm, bias 0.17°, band/mean 1.41. Two model corrections were required (per-grain parent selection and copy-mode boundary smoothing). The deck D is $0.22\,\mathrm{deg}^2/\mu\mathrm{m}$, half Lamb's height fit, because the inheritance path is 1.8× the height; the same D reproduces Lamb at the experiment's own 1.5 µm voxel. Open issues: the simulated grains are epitaxial substrate continuations, so Lamb's originating class is essentially absent, and the ⟨001⟩ fibre has no quantitative target.

1 Summary

- SROD in GP700 is a random-axis rotation walk with $D \approx 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$ against height and a Maxwell width. The authors' "fast first 30 µm, then constant" reading is $\sqrt{h}$.
- Both SPPARKS additive apps carry the model: each captured voxel takes its parent's orientation rotated by $\theta = \sqrt{D\,\mathrm{hop}}$ about a random axis. The misorientation cap and gradient-target rotation are removed.
- The GP700 Moser deck (`examples/gp700_moser/`) matches Lamb's SROD level, $\sqrt{h}$ shape and band width over the full 400 µm build (Table 2). Two corrections were needed: per-grain parent selection (mobility weighting between grains, not between the voxels of one grain, which had rotated grains toward ⟨001⟩) and `smooth_orientation copy` (the smoothing pass averaged neighbour quaternions, cancelling 60 % of the walk and widening the band to 1.8).
- Deck $D = 0.22\,\mathrm{deg}^2/\mu\mathrm{m}$ at 3 µm; the growth path is 1.8× the height. At 1.5 µm the same D stays on Lamb (bias 0.41°) with 10–18 % more accumulation per height (Table 3).
- Open: 83 of 88 filtered grains originate at the first remelt floor, so Lamb's originating class (> 500 grains, 5:1 over established) is essentially absent; the ⟨001⟩ fibre (0.19–0.24 within 15° of the build direction, twice random) has no quantitative experimental target.

2 Experimental methods (Lamb et al. 2023, Met. Trans. A 2026)

Material and process. CoNi-base superalloy GammaPrint-700, laser powder bed fusion at 176 W and 1137 mm/s, 80 µm spot ($D4\sigma$), 80 µm hatch, 30 µm layers, 90° interlayer rotation,

200 °C plate preheat. The 2026 paper supplies the thermophysical set used for its 3DThesis simulations: $k = 22.5\,\mathrm{W/(m\,K)}$, $\rho = 8650\,\mathrm{kg/m^3}$, $c_p = 600\,\mathrm{J/(kg\,K)}$ ($\alpha = 4.335 \times 10^{-6}\,\mathrm{m^2/s}$), absorptivity 0.9, liquidus 1646 K, solidus 1592 K.

3D dataset. TriBeam tomography (femtosecond-laser serial sectioning with EBSD on each slice), 1.5 µm in-plane step and 1.5 µm slice thickness, 385 slices, reconstructed into one HDF5 volume and segmented into grains at a 2° misorientation threshold before any SROD calculation.

SROD. For each grain the reference orientation sits near the origination point, "the lowest location of a grain's boundary along the build direction", offset one voxel layer (1.5 µm) upward so the reference is not a boundary pixel with a convoluted EBSD pattern; offsets up to 8 µm do not change the result. Two classes: *originating* grains, whose lowest point lies inside the volume, and *established* grains, cut by the dataset floor and referenced at the centroid of their footprint on the lowest slice. Grains shorter than 60 µm (two layers) or under 40 µm equivalent spherical diameter (ESD) are excluded as overtaken during competitive growth, leaving over 500 originating and over 100 established grains. Each grain is sliced along the build direction; each slice gives a SROD histogram fitted to a normal distribution and plotted against height above the origin (Fig. 5a) or above the dataset floor (Fig. 5c). The 2026 paper adds the cellular-scale finding used here: cells switch ⟨100⟩ growth variants at fusion lines rather than rotating continuously, so no gradient-directed rotation is modelled.

3 Analysis of the experimental data

Fig. 5a and 5c were digitized from 300 dpi renders of the paper (`docs/srod_random_walk.py`; curves in `docs/lamb2023_fig5{a,c}_srod.json`). The mean, 5th and 95th percentile curves were tested against a rotation walk: each increment is a small rotation of angle θ about a random axis; after n independent increments the misorientation from the origin has

$$\mathrm{RMS}^2 = n\,\theta^2 = D\,d \tag{1}$$

and a Maxwell distribution (χ with three degrees of freedom), for which $(p_{95} - p_5)/\mathrm{mean} = 1.38$ at every distance. $D = 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$ fitted on the originating grains predicts the established grains without refitting; the measured width ratio is 1.3–1.5 at every height; the apparent two-regime shape is the square root (Fig. 1). A constant per-hop angle is not required, only independent increments with finite variance, so D can carry an undercooling dependence without changing the $\sqrt{h}$ law.

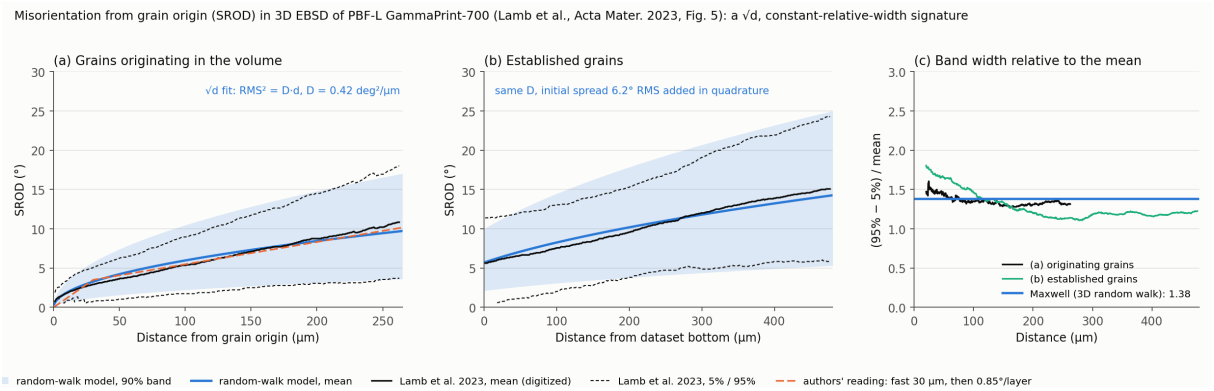


Figure 1: Lamb 2023 Fig. 5 digitized. Left: mean SROD of originating grains with $\mathrm{RMS}^2 = Dh$, $D = 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$; the same D drawn through the established grains. Right: $(p_{95} - p_5)/\mathrm{mean}$ against height, flat at 1.3–1.5 around the Maxwell value 1.38.

4 Simulation methods

Code. SPPARKS, branch SURGE-kokkos, app `additive/texture/kk` (Kokkos/HIP on an AMD gfx1151 APU), one MPI rank. The CPU app `additive/texture` carries the same model and is the correctness reference; the temperature field is deterministic and bit-identical between the two (`examples/am_validation/`).

Thermal source. Moser Green's-function solution with the GP700 constants (`setup_temperature_source moser 176 0.9 22.5 4.335e-6 473 600 20e-6 20e-6 10e-6`), evaluated on the device for every site each step. Scan lines with `hatch 80e-6 N serpentine pause 8.8e-3`, the pause being the real 10 mm track length over the scan speed; the fast-forward predictor skips idle gaps. Single-hatch check: pool 147 × 69 µm, thermal gradient 1×10^7–2.9×10^7 K/m at the fusion boundary (paper: 3×10^7 K/m).

Build. Box-per-layer stitch continuation (`run_layers.sh`). Domain 801 µm square (267 voxels at 3 µm), 10 hatches at $40 + 80k$ µm with lines 100 µm past both edges, 12 layers of 30 µm, 90° rotation per layer, each box a 180 µm window below the laser plane; each layer starts from the preheat (the recoat interval). Substrate: 150 µm of 50 µm Voronoi grains with uniform random orientations (`make_substrate.py`); five overlapping hatches remelt 80–100 µm, so the analysable volume begins at 105 µm. Timestep 4 µs, twice the $\Delta x/V(T_l - T_s)$ guard; 2 and 4 µs agree within seed noise. The 1.5 µm run uses 400 voxels over 600 µm, 7 hatches, 8 layers, $\Delta t = 2$ µs.

Solidification kinetics. Front velocity is the 304L polynomial ×3.5 (front at 25–35 K undercooling, about 0.56 m/s, keeping up with the 1.137 m/s scan); nucleation density $1.5 \times 10^{13}/\mathrm{m}^3$, critical undercooling 15 K, deviation 2 K. Parent selection per grain: candidate neighbours are grouped by grain, each grain weighted by the mobility $1 + c_2 \cos(c_3\theta)$ of its sign-aligned mean orientation against the local thermal gradient ($c_2 = 2.8$, $c_3 = 2.5$, negative clamped to zero), then the parent voxel is drawn uniformly inside the chosen grain.

Misorientation. `misorientation_diffusion 2.2e5 30`: every captured voxel takes its parent's orientation rotated by $\theta = \sqrt{D\,\mathrm{hop}}$ about a random axis (hop = lattice distance to the parent in m, D in $\mathrm{deg}^2/\mathrm{m}$); nucleus shells rotate relative to the seed by the same rule; a rotation past 30° would start a new grain (never reached). No cap, no gradient target, constant D.

Boundary smoothing. One greedy steepest-descent pass per captured voxel (`smooth_greedy_multiproposal on`, nrefine 1) the step after capture, and `smooth_orientation copy`: a voxel that changes grain takes one random neighbour's orientation from that grain. The default `average` takes the sign-aligned mean.

Kokkos kernel corrections made during this work, all now at CPU parity: race-free multicolour neighbour reads (they were x-index ordered), front advance only beside solid, nrefine passes honoured, zero-gradient mobility, and the `laser_path` time anchor after `reset_time`.

5 Simulation analysis methods

`gp700_analysis.py` on the spparks-micro-analysis tools: read the last stitch frame; solid means `i2 == 3`, so unremelted substrate is excluded; crop to the central 400 µm laterally (300 µm on the proxy), optional ceiling; relabel grains as connected components of spin; grain volume, ESD and extents; Lamb's filters (height ≥ 60 µm, ESD ≥ 40 µm); origin = the grain's lowest solid voxel, whose quaternion is the reference (exact in the simulation, so no offset); grains crossing the analysis floor are established and referenced at their floor-slice centroid. SROD is

the symmetry-reduced disorientation of every voxel from its grain's reference, binned by height above the origin or floor in 10 µm steps and, separately, by straight-line distance, with mean, p_5, p_{95} and RMS per bin. D-fit: $\mathrm{RMS}^2 = Dh$ through the origin, volume weighted. Score: RMS of (simulated mean − Lamb Fig. 5a mean) over 25–240 µm, plus bias, $(p_{95} - p_5)$/mean, the fraction of solid within 15° of ⟨001⟩ parallel to the build direction, and grain counts. Sobol designs (scipy `qmc`) on a 600 µm, 8-layer proxy at 6 min per box were used for calibration.

6 Results

6.1 Calibration path

Table 1: Full builds against Lamb Fig. 5a (all-originating SROD, central 400 µm crop).

build	change	RMS / bias vs Lamb	band/mean	⟨001⟩ within 15°
v2	first full build, D 0.42, cap removed	2.2° / 3.6° (linear)	1.4–1.8	0.26
v4	per-grain parent selection	2.9° low	1.8	0.11
v5	Sobol: D 1.15, nucl 1.5×10^{13}, t_c 15, c_2 2.8	0.59° / −0.16°	1.79	0.34
v6	50 µm Voronoi substrate, nrefine 1, D 0.40	2.2° / −1.8°	1.85	0.16
v7a	D 0.80	0.90° / 0.24°	1.85	0.20
v8	**`smooth_orientation copy`, D 0.22**	**0.49° / 0.17°**	**1.41**	0.19
v9	c_2 9, nucl 2×10^{13}	0.52° / 0.02°	1.45	0.22

v2's linear growth was selection drift: mobility weighting among the randomly rotated voxels of one grain rotated grains toward ⟨001⟩ with height, fixed by per-grain selection. v4 to v7 then retained only 0.3–0.45 of the input D against height in every configuration while the single-grain walk deck retained 1.2. The smoothing pass replaced each reassigned voxel's orientation with the average of its new neighbours, cancelling the fresh rotations and narrowing the intragranular distribution (band 1.8). Copy mode restored retention to 1.6–1.9 and the band to 1.4 on the proxy at $D = 0.42$; scaling that curve onto Lamb gave $D = 0.22$.

6.2 v8 against Lamb

Table 2: v8 (copy mode, D 0.22 deg²/µm, Voronoi substrate, nrefine 1, nucleation 1.5×10^{13}/m³, t_c 15 K, c_2 2.8, Δt 4 µs).

	v8	Lamb 2023
SROD RMS deviation, 25–240 µm	0.49°	–
mean bias	0.17°	–
mean SROD at 54 / 204 / 402 µm	4.4 / 8.5 / 12.3°	3.8 / 8.8 / – (data end at 262 µm)
$(p_{95} - p_5)$/mean	1.41	1.4
D-fit vs height / straight line	0.41 / 0.40 deg²/µm	0.42
grains passing filters	86 (400 µm crop)	> 600 (full volume)
ESD p_{10} / p_{50} / p_{90}	73 / 151 / 182 µm	–
lateral extents, median	114 × 99 µm	–
⟨001⟩ within 15° of BD	0.19 (random 0.09)	qualitative ⟨001⟩ fibre

6.3 Resolution: 1.5 µm versus 3 µm

The per-hop rule $\theta = \sqrt{D\,\mathrm{hop}}$ is resolution independent to first order; the finer lattice resolves a slightly longer growth path per unit height, and an exact match at 1.5 µm would take $D \approx 0.19$.

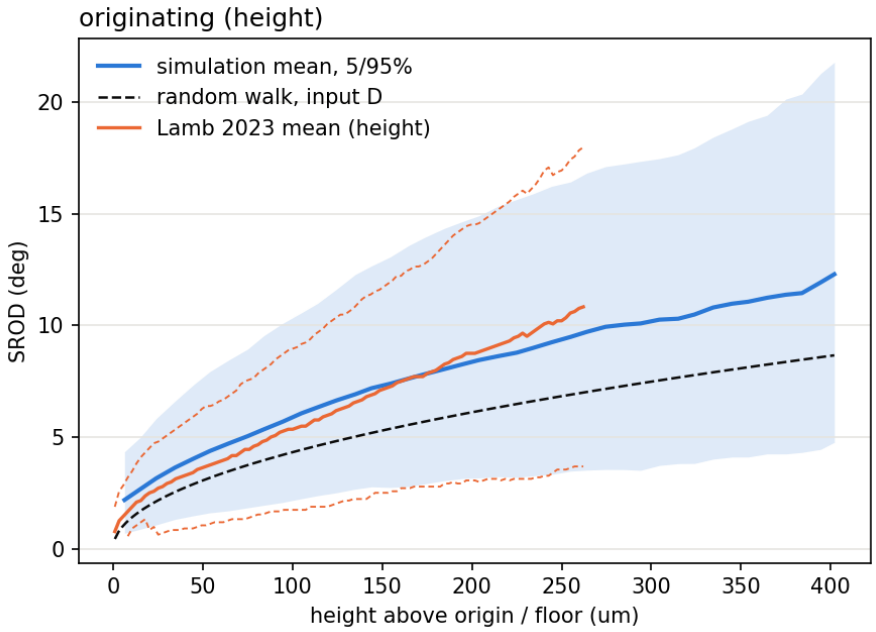


Figure 2: v8, all-originating SROD against height above the origin (central 400 µm crop, floor 100 µm): simulated mean with the 5/95 band (blue) against Lamb 2023 mean and percentiles (orange). The dashed line is the input D drawn against height; the simulated curve runs above it by the path factor.

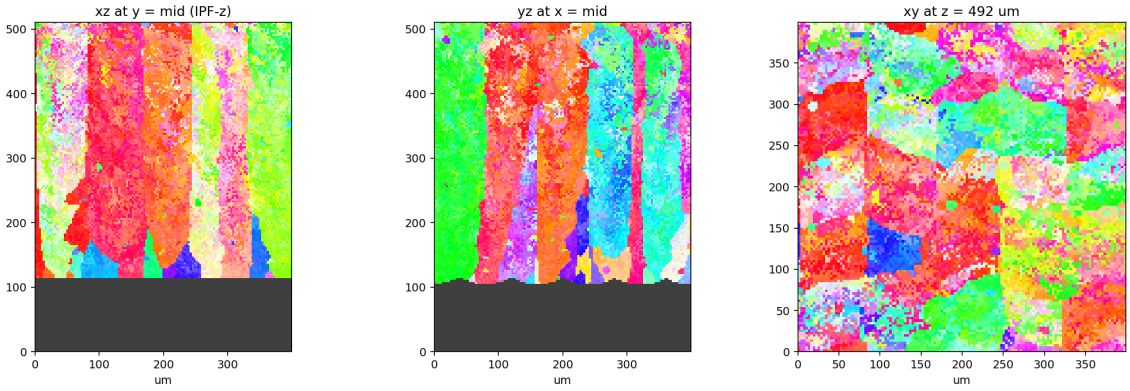


Figure 3: v8 IPF-Z colouring: xz and yz sections through the crop centre, and the xy section at 492 µm. Dark grey is the unremelted substrate.

Table 3: Same deck, geometry and seed on the 600 µm proxy.

	3 µm	1.5 µm
SROD RMS / bias vs Lamb	0.44° / 0.03°	0.49° / 0.41°
band/mean	1.41	1.43
retention (D-fit ÷ input), height / straight line	1.83 / 1.73	2.17 / 2.35
⟨001⟩ within 15°	0.21	0.20
ESD p_{10} / p_{50} / p_{90} (µm)	68 / 123 / 159	66 / 122 / 150
wall time per box	6 min	47 min

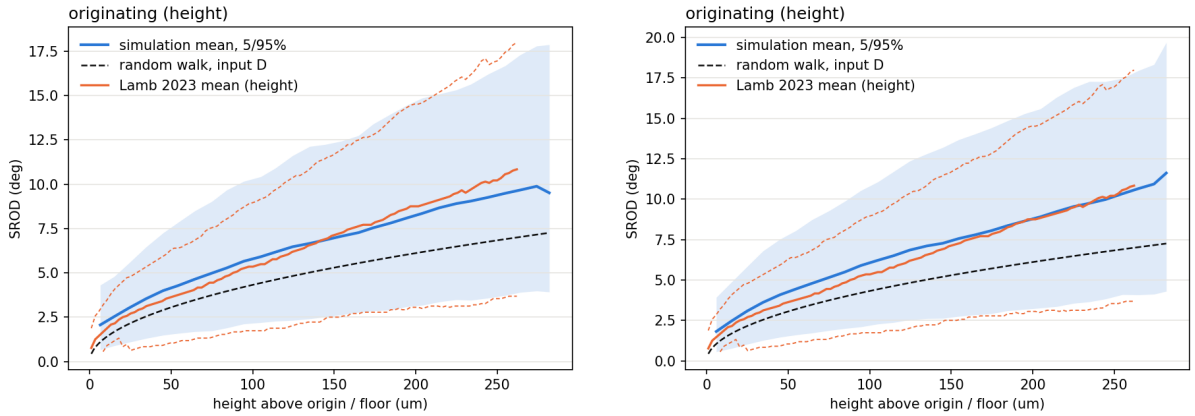


Figure 4: Left: 3 µm reference. Right: 1.5 µm, Lamb's voxel size. Bin-by-bin agreement of 0.1–0.4° to 200 µm, 7–10 % higher at 1.5 µm beyond 245 µm.

6.4 Crop sensitivity and grain origins

Analysing v8 on the full 801 µm domain, the central 400 µm, the central 200 µm, or with the never-remelted top layer removed changes the SROD bias by under 0.2° and the band by under 0.03; only the fibre fraction moves, and it cannot be measured on 24 grains. Of the 88 filtered grains in the standard crop, 83 originate at the layer-0 remelt floor (105–130 µm, 99 % of the filtered volume) and 3 above 160 µm. The columnar grains are epitaxial continuations of substrate grains, so the simulated "originating" SROD is an established-grain SROD referenced at the remelt floor. Because Lamb's two classes share the same D, the walk comparison is unaffected, but the model does not generate Lamb's population of grains that originate mid-build.

6.5 Texture

A Sobol study over $c_2 \in [2, 8]$ and nucleation 5×10^{12}–5×10^{13} with D fixed gives $\langle 001 \rangle = -0.042 + 0.0125\, c_2 + 0.0073 \ln(\text{nucl})$ ($R^2 = 0.72$), flat at 0.22 below $c_2 \approx 5$, with the SROD score unaffected throughout. On the full build $c_2 = 9$ gives 0.22, rising from 0.16 to 0.24 with height, while $c_2 = 2.8$ saturates at 0.19. The walk spreads orientations out of the 15° cone (13° RMS over 400 µm) about as fast as selection fills it. Neither paper quantifies the fibre.

6.6 Per-grain orientation paths and the grain-level walk

Lamb's Fig. 4 follows individual grains: an IPF map, a SROD map, the mean SROD per layer, and every voxel of the grain on the build-direction inverse pole figure, where a red arrow marks the rotation path from bottom to top and one grain (G4) splits into two clusters. `grain_paths.py` reproduces that layout. For v8's four tallest grains the mean orientation drifts only 3–6° from bottom to top while the voxels spread 12° (RMS) about the top-slab mean: the per-voxel walk lets the lineages of one grain diverge independently, so the cloud expands isotropically and its mean stays put, whereas Lamb's grains rotate coherently along a path.

`misorientation_grain_walk` D_g adds a coherent walk: every voxel of grain g crossing lattice level $k \to k+1$ receives the same rotation, of angle $\sqrt{D_g\, \Delta x}$ about an axis hashed from (g, k), so the grain's coherent rotation is a function of (g, z) only, path independent and without per-grain state; the per-voxel walk adds intragranular spread on top. Table 4 compares three proxy runs with the v8 deck otherwise (Fig. 5).

All three runs sit on Lamb's population curve with the same band ratio, so the population SROD statistics cannot distinguish coherent from per-voxel accumulation: each grain's coherent walk is one realisation and the Maxwell width appears across grains either way. The per-grain analysis does distinguish them, and the ratio $D_g : D_v$ sets drift against spread. Two further

Table 4: Per-voxel, mixed and coherent walks on the 3 µm proxy (600 µm, 8 layers); per-grain rows are for the four tallest filtered grains.

	voxel only (D_v 0.22)	mixed (D_g 0.30 + D_v 0.065)	grain only (D_g 0.42)
SROD RMS / bias vs Lamb	0.44° / 0.03°	0.54° / 0.34°	0.47° / 0.04°
band/mean (population)	1.41	1.46	1.48
D-fit vs height (deg^2/µm)	0.40	0.43	0.38
same-grain z-neighbour disorientation, median / p_{90}	3.8 / 12.5°	2.3 / 6.7°	1.1 / 1.7°
drift of the mean orientation, bottom → top	0.7–6.3°	6.5–8.8°	5.3–11.3°
spread about the top-slab mean (RMS)	7.6–11.7°	4.2–7.1°	1.4–2.0°
maximum SROD in the grain	20–28°	16–19°	8–12°

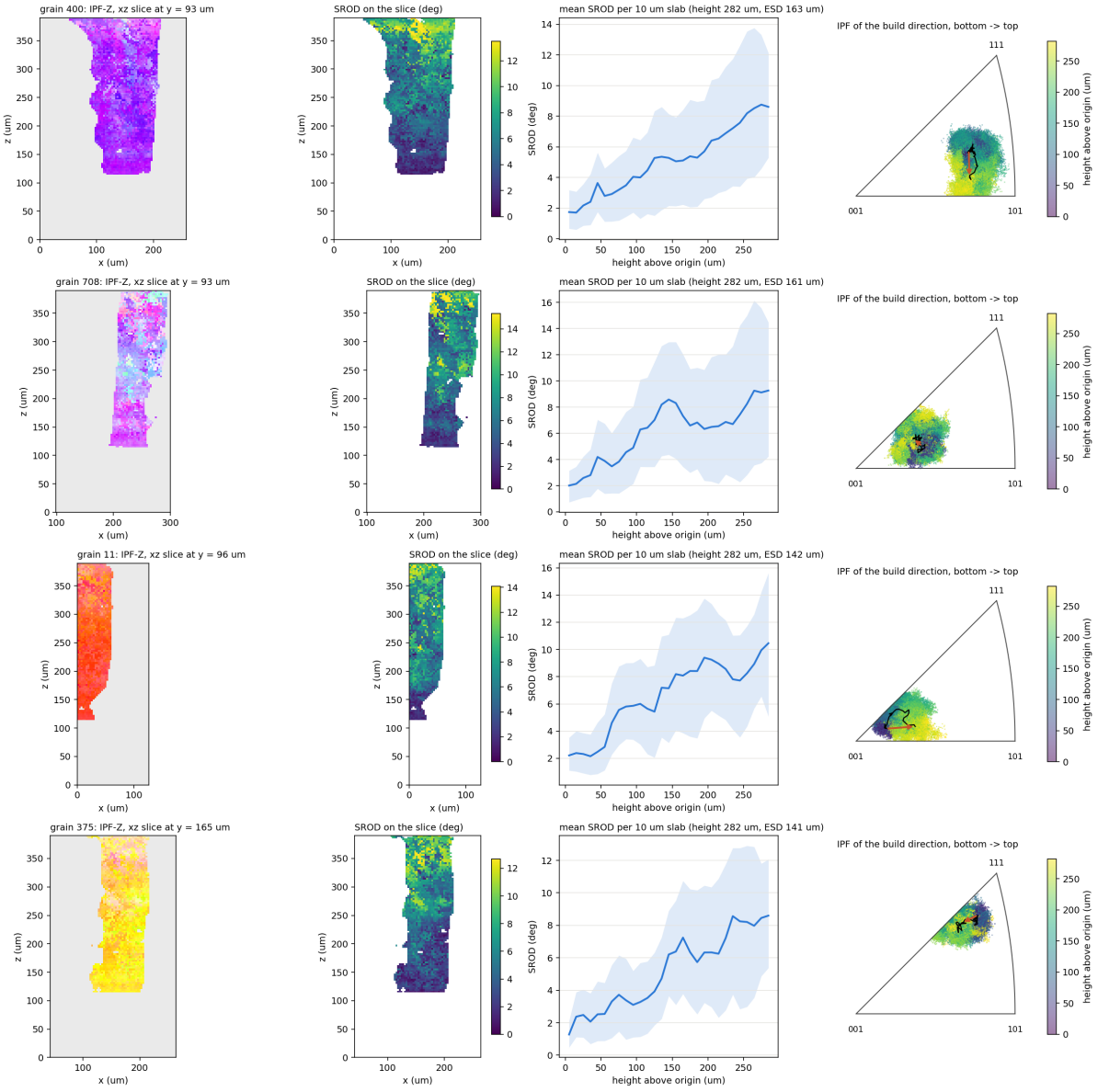


Figure 5: Mixed grain + voxel walk on the proxy, four tallest grains: IPF-Z and SROD maps on the xz slice through the grain, mean SROD per 10 µm slab with the 5/95 band, and every voxel on the build-direction inverse pole figure coloured by height (black: per-slab mean path; orange: bottom to top). The voxel-only and grain-only versions are `grain_paths_voxel_only.png` and `grain_paths_grain_only.png` in the same directory.

observations from the grain-only run: a memoryless walk makes individual SROD(h) curves non-monotonic (one grain reaches 8° at 150 µm and returns to 2° at 240 µm), whereas Lamb's G1–G4 rise steadily to 20–25°, which points to persistent, correlated-axis rotation (lattice bending) rather than a memoryless walk; and the per-level increment (1.1° at 3 µm) shows as horizontal bands in the IPF map. Lamb's within-slice widths are not published numerically, so the split is constrained only to drift $\gtrsim$ spread; the mixed setting is the working choice.

Persistent rotation. With `misorientation_grain_walk` D_g ρ the per-level rotation vector is an Ornstein–Uhlenbeck sequence in the level index (correlation height $\approx \Delta x/(1-\rho)$), still hashed from (g, k) and evaluated from level 0. A fixed per-level angle with a persistent axis made every grain rotate at the same rate and collapsed the population band (0.77 at $\rho = 0.9$), so the increment is a Gaussian rotation vector of RMS $\sqrt{D_g \Delta x}$; the accumulated rotation across grains is then Maxwell in both regimes. Table 5 gives the second round on the proxy, with D_g scaled for a long-range population D near 0.42 (the empirical amplification is $\approx 15\times$ at $\rho = 0.9$ and $\approx 32\times$ at $\rho = 0.97$, below the AR(1) values 19 and 66).

Table 5: Persistent (OU) grain walk on the 3 µm proxy; per-grain rows for the four tallest filtered grains.

	grain only, ρ 0.9 (D_g 0.029)	mixed, ρ 0.9 (D_g 0.018 + D_v 0.065)	grain only, ρ 0.97 (D_g 0.0115)
SROD RMS / bias vs Lamb	0.77° / 0.64°	0.42° / 0.11°	0.59° / −0.50°
band/mean (population)	1.35	1.44	1.66
mean SROD at 6 / 45 / 124 / 244 µm	0.5 / 3.2 / 7.3 / 10.9°	1.1 / 3.3 / 6.8 / 9.3°	0.3 / 2.4 / 6.1 / 9.4°
drift of the mean, bottom → top	4.0–11.3°	1.9–8.6°	3.0–14.1°
spread about the top-slab mean (RMS)	0.5–1.1°	5.0–6.3°	0.5–1.0°
maximum SROD ÷ top-slab mean	1.0–1.3	≈ 2	1.0–1.2

Persistence makes individual grains rise monotonically along a directed IPF path with smooth intragranular gradients, keeps the population band at Lamb's value, and turns the population curve ballistic below the correlation height (mean SROD $\propto h$ up to ∼45 µm at $\rho = 0.9$, ∼100 µm at $\rho = 0.97$, then $\sqrt{h}$), which is the "fast first layer" shape Lamb describe. The mixed $\rho = 0.9$ run sits on Lamb (0.42°, 0.11°, band 1.44) and is the working candidate for the next full build. Two observables would fix the remaining freedom: Lamb's within-slice width sets $D_g : D_v$, and their population curve below 30 µm sets ρ; neither is resolved by the published figures.

7 Findings and open issues

1. SROD in GP700 is a rotation walk with $D \approx 0.42\,\mathrm{deg}^2/\mu\mathrm{m}$ against height and a Maxwell width; both grain classes obey it.
2. The model reproduces level, shape and width once orientation is inherited from one parent voxel and never averaged. Per-grain parent selection and copy-mode smoothing are both required; averaging in either place damps the walk and narrows the band.
3. Deck D is $0.22\,\mathrm{deg}^2/\mu\mathrm{m}$ at 3 µm and about 0.19 at 1.5 µm: the walk runs along the inheritance path, 1.8× the height in this remelt geometry. Whether the real path factor is that large is an open physical question; D should be read per unit path length.
4. Grain classes do not match. No surviving grains originate mid-build at nucleation $1.5 \times 10^{13}/\mathrm{m}^3$ with Lamb's filters. Lamb's 5:1 originating-to-established ratio implies nucleation or lateral overgrowth the model does not produce at these settings.
5. Individual grains in the experiment rotate coherently along a path; the per-voxel walk alone spreads without drifting. The grain-level walk (`misorientation_grain_walk`) reproduces the drift with the same population statistics, and a persistent (Ornstein–Uhlenbeck) form of it reproduces the steady per-grain rise and the ballistic first layer while staying on the population curve.
6. The ⟨001⟩ fibre is twice random and moves only weakly with the mobility contrast; there is no measured target.

7. Not yet tested: the undercooling dependence of D ($\alpha > 0$); the single-hatch check deck is hot enough that α barely acts.

8 Reproduction

```
. /home/tron/spack/share/spack/setup-env.sh && spack env activate spparks-dev
# binary: build-verify/spparks (commit da2fad29 or later)
E=examples/gp700_moser; R=/home/tron/gp700_runs
$E/run_layers.sh build-verify/spparks $R/wide12_v8 12 12345 267 10 4e-6 $R/
    substrate_800um_50um.st
/home/tron/miniconda3/envs/WORK/bin/python $E/gp700_analysis.py $R/wide12_v8/gp700.st
    \
    --out $R/wide12_v8/analysis_floor100 --floor-um 100 --crop 200 600 200 600 --D 2.2
        e5
DX=1.5e-6 $E/run_layers.sh build-verify/spparks $R/dx1p5_v8 8 12345 400 7 2e-6 $R/
    substrate_600um_50um_dx1p5.st
```

Runs live in `/home/tron/gp700_runs/` (wide12_v2 to v9, dx1p5_v8, dx3_v8_ref, sobol1/2, sobol_fibre1/2, smooth_*). Key commits: 8325eb71 (cap), fa8930f3 (diffusion model), ad23c1a5 (per-grain selection), 1a9712b3 (kk kernel parity), da2fad29 (`smooth_orientation`), a0fb1011 (v8 deck), a3a66f41 (voxel-size driver), b6e4ba5f / 47546146 (`misorientation_grain_walk`, persistence). This file is the reference version; `docs/SROD_ANALYSIS_REPORT.md` and the published HTML page mirror it.

Sources. J. Lamb et al., Acta Materialia (2023), 3D-EBSD and SROD of PBF-L GammaPrint-700; J. Lamb et al., Metallurgical and Materials Transactions A (2026), cellular structure and 3DThesis parameters.